GSS / GAME DESIGN AND SELECTION RATIONALE

A puzzle for discovering rare disease biology.

GREGoR Puzzles | 10-round collection | 10 October 2026

This game uses an image jigsaw to introduce a gene, what it does, a condition associated with it, and that condition's inheritance pattern. The scientific information opens gradually as the player assembles the illustration.

The key idea

The puzzle is about assembling an explanation. The image shows a biological process, not a patient. Gene and condition names are the final two discoveries.

What happens in one round

1. Choose a round

Pick any of 10 topics. Each round contains 12 interlocking image pieces.

2. Build and reveal

Place pieces by mouse or by tapping a piece and its board position. Clues unlock at 2, 4, 6, and 8 pieces.

3. Connect the science

Two questions link the diagram to mechanism and inheritance. Explanations support retrying an answer.

4. Discover the names

The gene appears at 11 pieces and the condition at 12. Recognizing the gene can finish the picture, but both questions still count.

5. Read and continue

Review phenotype summaries, variant classes, scientific references, and a community resource. Progress saves separately for each round.

Practice mode is untimed. Optional timed mode continues across reloads and time away until the picture and both questions are complete. There is no shared leaderboard.

SELECTION APPROACH

Why these 10?

I selected a small teaching collection from the gene-condition pairs you supplied, keeping their relative order. I looked for distinct biological processes, clear illustrations, well-supported public explanations, and useful comparisons across rounds.

Educational selection, not a clinical ranking

The supplied gene-condition list does not establish which conditions are most important to GSS, most frequent among families, or highest priority for ASHG. Matt's feedback asked for that prioritization; this collection is a proposal for him to review, not a claim that clinical priority has been established.

Breadth: The set spans membrane conductance, cell communication, lysosomal transport, ribosome maturation, major and minor splicing, developmental signaling, immune signaling, and a repeat expansion.

Comparisons: CLCN1 and CACNA1A show channels in different tissues. RNU4-2 and RNU4ATAC show two RNA-splicing systems. PTPN11 and STING1 show signaling in different biological settings.

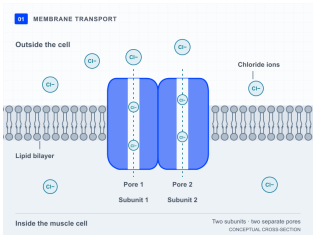
A workable first release: Each topic can be introduced with one conceptual picture, six clue milestones, and two questions. That keeps the initial game coherent and reviewable.

The collection at a glance

ROUND / GENE	CONDITION
01 / CLCN1	Recessive myotonia congenita
02 / GJB2	Recessive nonsyndromic hearing loss 1A
03 / SLC29A3	Histiocytosis-lymphadenopathy plus syndrome
04 / SBDS	Shwachman-Diamond syndrome 1
05 / RNU4-2	ReNU syndrome
06 / CACNA1A	Developmental and epileptic encephalopathy 42
07 / PTPN11	Noonan syndrome 1
08 / RNU4ATAC	Roifman syndrome
09 / STING1	STING-associated vasculopathy with onset in infancy
10 / FGF14	Spinocerebellar ataxia 27B

ROUNDS 01-02

Two ways that cells use channels



Membrane transport

01 CLCN1

Recessive myotonia congenita

Autosomal recessive | OMIM 255700

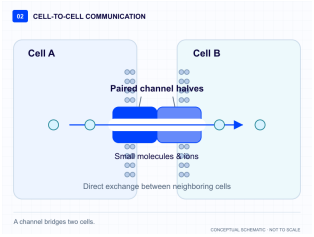
Features introduced: Myotonia; Muscle stiffness; Warm-up effect.

Why this level: The first supplied pair gives players a direct connection between a muscle-cell chloride channel and delayed muscle relaxation. The two-pore membrane illustration offers a concrete introduction before the collection moves into RNA and signaling.

Player takeaway: A gene encodes a protein with a cellular role; altered conductance can change muscle behavior.

Keep precise: CLCN1 has dominant and recessive associations. This level is specifically the recessive condition.

Sources: OMIM 255700 | MedlinePlus · genetics overview | GeneReviews · clinical reference



Cell communication

02 GJB2

Autosomal recessive nonsyndromic hearing loss 1A

Autosomal recessive | OMIM 220290

Features introduced: Sensorineural hearing loss; Often present early; Variable degree.

Why this level: GJB2 broadens the introduction from one cell membrane to direct communication between cells. It supports a legible paired-channel image and a recognizable sensory example while keeping the inheritance pattern recessive.

Player takeaway: Gap junctions connect neighboring cells and allow ions and small molecules to pass between them.

Keep precise: Use respectful descriptions of hearing differences. This level is the nonsyndromic recessive association, not all GJB2 conditions.

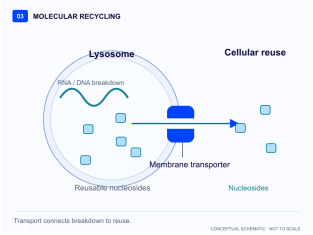
Sources: OMIM 220290 | MedlinePlus · GJB2 | GeneReviews · recessive hearing loss

Why these make a useful introduction

Both diagrams show membrane-associated structures, but the biological jobs differ. CLCN1 concerns electrical balance within muscle cells; GJB2 concerns communication between neighboring cells. The familiar visual format supports comparison rather than memorizing two gene names.

ROUNDS 03-04

Recycling molecules and building machinery



Lysosomal transport

03 SLC29A3

Histiocytosis-lymphadenopathy plus syndrome

Autosomal recessive | OMIM 602782

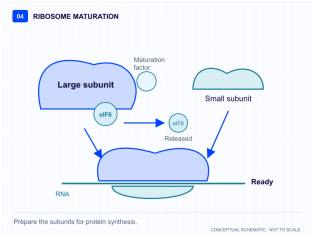
Features introduced: Histiocytosis; Lymphadenopathy; Skin changes.

Why this level: SLC29A3 introduces lysosomal nucleoside transport, a process absent from the two channel levels. It also provides a useful example of explaining a gene function while acknowledging that the full chain to every clinical feature remains under study.

Player takeaway: A transporter helps handle reusable molecular building blocks inside cells.

Keep precise: The illustration explains a simplified transport role. It does not establish a complete mechanism for histiocytosis.

Sources: OMIM 602782 | MedlinePlus · condition | MedlinePlus · SLC29A3



Ribosome assembly

04 SBDS

Shwachman-Diamond syndrome 1

Autosomal recessive | OMIM 260400

Features introduced: Pancreatic insufficiency; Neutropenia; Skeletal differences.

Why this level: SBDS adds ribosome maturation, making the collection about more than channels and signaling. Its assembly story is well suited to a puzzle: release a blocking factor, then allow the ribosomal subunits to join.

Player takeaway: A maturation factor can be essential to building the machinery for protein synthesis.

Keep precise: The conceptual drawing separates released eIF6 from the assembled ribosome. SBDS is one of several related disease genes.

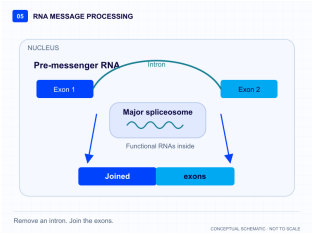
Sources: OMIM 260400 | MedlinePlus · SBDS | MedlinePlus · condition

What these add to the collection

These rounds shift the focus inside the cell. One concerns moving reusable building blocks; the other concerns preparing the machinery for protein synthesis. They broaden the set beyond a collection of channel disorders.

ROUNDS 05-06

Functional RNA and neural signaling



Major RNA splicing

05 RNU4-2

ReNU syndrome

Autosomal dominant | OMIM 620851

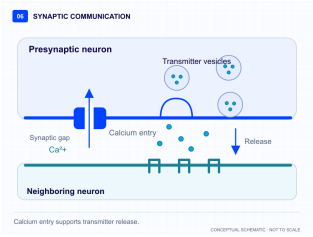
Features introduced: Developmental delay; Hypotonia; Speech and language differences.

Why this level: RNU4-2 was an explicit priority in the meeting notes. It is an important teaching counterexample to the assumption that every disease-associated gene encodes a protein. It also creates a direct comparison with RNU4ATAC later in the collection.

Player takeaway: A change in an RNA gene can alter splicing without changing a protein-coding sequence.

Keep precise: This round concerns the dominant ReNU association. Do not generalize that pattern to every RNU4-2 variant or association.

Sources: OMIM 620851 | Chen et al. · Nature 2024 | GeneReviews · ReNU syndrome



Synaptic communication

06 CACNA1A

Developmental and epileptic encephalopathy 42

Autosomal dominant | OMIM 617106

Features introduced: Epilepsy; Developmental delay; Hypotonia or ataxia.

Why this level: CACNA1A lets players transfer the channel idea from skeletal muscle to neurons. Calcium entry and neurotransmitter release produce a distinct synapse image and demonstrate why biological context matters even when two genes both encode channels.

Player takeaway: Ion-channel activity helps couple an electrical signal to communication between neurons.

Keep precise: The selected condition is DEE42. CACNA1A has multiple associations and variant effects; a single gain- or loss-of-function story is insufficient.

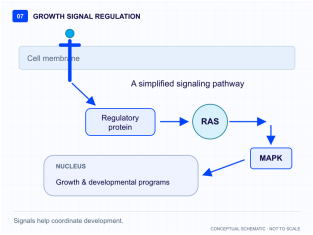
Sources: OMIM 617106 | MedlinePlus · CACNA1A | Jiang et al. · channel effects in DEE

What this pairing makes visible

A functional RNA and a protein channel can both be associated with neurological features through very different processes. The levels emphasize gene function and the specific gene-condition relationship rather than implying that one feature uniquely identifies a gene.

ROUNDS 07-08

Signal regulation and a second RNA system



RAS / MAPK signaling

07 PTPN11

Noonan syndrome 1

Autosomal dominant | OMIM 163950

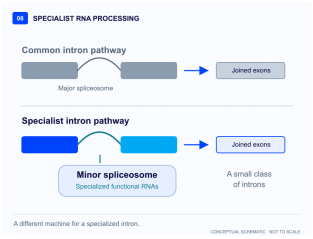
Features introduced: Short stature; Heart differences; Bleeding tendency.

Why this level: PTPN11 provides an accessible example of a signaling regulator in the RAS/MAPK pathway. It adds a developmental syndrome and helps show that a disease mechanism can involve altered signal regulation rather than simply an absent protein.

Player takeaway: A regulatory protein helps control signals that guide growth and development.

Keep precise: This is PTPN11-related Noonan syndrome 1. Noonan syndrome is genetically heterogeneous.

Sources: OMIM 163950 | MedlinePlus · PTPN11 | MedlinePlus · Noonan syndrome



Minor RNA splicing

08 RNU4ATAC

Roifman syndrome

Autosomal recessive | OMIM 616651

Features introduced: Skeletal dysplasia; Immunodeficiency; Retinal dystrophy.

Why this level: RNU4ATAC was another meeting priority. Roifman syndrome supplies a second functional-RNA example with a different spliceosome and recessive inheritance. The comparison is intentional: similar-looking gene names do not imply identical biology or inheritance.

Player takeaway: The minor spliceosome processes a small, specialized class of introns.

Keep precise: This level covers Roifman syndrome, not the separate RNU4ATAC-related primordial dwarfism condition also listed in the table.

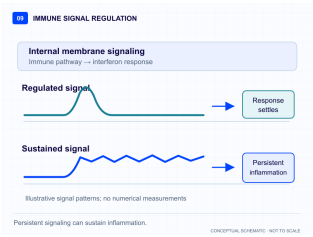
Sources: OMIM 616651 | Merico et al. · Nature Communications 2015 | GeneReviews · RNU4atac-opathy

Why include both RNA genes?

RNU4-2 and RNU4ATAC are not redundant teaching examples. Together they explain that a gene may encode a functional RNA, that more than one spliceosome exists, and that inheritance belongs to the specific gene-condition relationship.

ROUNDS 09-10

Immune signaling and a repeat expansion



Innate immune signaling

09 STING1

STING-associated vasculopathy with onset in infancy

Autosomal dominant | OMIM 615934

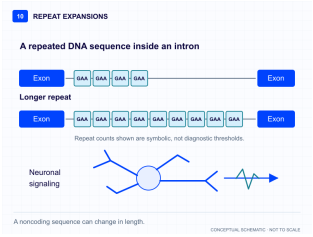
Features introduced: Skin inflammation; Blood-vessel involvement; Interstitial lung disease.

Why this level: STING1 adds an immune pathway and a clear sustained-signal mechanism. It contrasts with growth signaling in PTPN11 and with impaired transport or assembly in earlier rounds, expanding the idea of what a disease-causing change can do.

Player takeaway: Persistent interferon signaling can drive inflammation.

Keep precise: The signal traces are illustrative, not measurements. This round focuses on STING1 gain-of-function associated with SAVI.

Sources: OMIM 615934 | MedlinePlus · SAVI | MedlinePlus · STING1



Repeat expansions

10 FGF14

Spinocerebellar ataxia 27B

Autosomal dominant | OMIM 620174

Features introduced: Gait ataxia; Episodic symptoms; Downbeat nystagmus.

Why this level: FGF14 ends the set with a different kind of genetic change: a GAA repeat expansion inside an intron. It introduces a noncoding mechanism and an often adult-onset condition, broadening a set otherwise weighted toward early-onset disorders.

Player takeaway: A DNA sequence can gain repeat units; noncoding changes can have biological consequences.

Keep precise: This is SCA27B, distinct from other FGF14 conditions. The drawing does not depict diagnostic repeat thresholds.

Sources: OMIM 620174 | GeneReviews · GAA-FGF14-related ataxia | Pellerin et al. · NEJM repeat discovery

Why end with FGF14?

It broadens the learner's model of a variant. The earlier rounds introduce altered protein or RNA function; the final round adds repeat length and an intronic location. This is a teaching sequence, not a statement that repeat expansions are inherently more difficult.

REVIEW AND NEXT DECISIONS

What this proposal establishes - and what still needs input.

The game is implemented as an introductory learning experience. The mechanisms and phenotypes have linked sources, but the content has not yet been approved by Matt or the GSS team. Playing successfully is not evidence of clinical competence.

Questions for Matt and the GSS team

Does this represent the program?

Which of these gene-condition pairs best reflects GSS discoveries, the audience, and the story the group wants to tell at ASHG?

Should any level be replaced?

Use cohort relevance, GSS contribution, a foundation partnership, or a specific educational need to guide changes. Those inputs were not available in the supplied list.

Are the scientific shortcuts acceptable?

Vet the mechanism, inheritance, phenotype summaries, general variant classes, and every illustration. In particular, retain condition-specific distinctions for CLCN1, CACNA1A, RNU4-2, RNU4ATAC, and FGF14.

Does the experience teach enough?

The jigsaw motivates exploration; the learning comes from its clues, questions, and explanations. A high-school introduction, a college course, and counselor exam preparation would require different depth and evaluation.

Source and data boundaries

Each level links to the supplied OMIM condition identifier. OMIM text could not be retrieved directly; the phenotype prose is an original summary supported by the other linked references, including MedlinePlus Genetics, GeneReviews, and primary research. It should not be described as an OMIM phenotype extraction.

Only gene-condition relationships were used. No family records, patient-specific phenotypes, individual variants, or sequencing metadata are in the game or this PDF. The slim input list did not include a site field for independently verifying Stanford membership. General variant descriptions are not family findings.

FAM177A1 was discussed in the meeting but was not in the supplied selection used for these 10 rounds. Roifman syndrome and the separate RNU4ATAC-related primordial dwarfism condition should remain distinct levels if both are later included. Repeated rows for the same gene-condition pair do not create extra levels.

Development repository | The in-game Content review page collects all ten levels, references, and answers.